

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 19

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Images / maps of large areas; land-use change, deforestation, flooding • Vegetation density and surface temperature • Wind velocity, sea-surface altitude, wave height, oil pollution	3
01.2	• Microwaves pass through clouds (unlike visible light)	1
01.3	• Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected • No need for researchers to visit inaccessible areas, reducing cost and disturbance; data are standardised across the whole area	2
01.4	<i>Any 4 from:</i> • Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night • Each tiger has a unique pattern of facial / body stripes • Images are compared with the database so individuals are recognised rather than counted twice • The number of different individuals recorded gives a minimum population; repeated captures give territory size, movements, lifespans and social groupings • Limitation: only works for species with unique visible features	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6.	
02.1	• Energy: low energy-density renewable (solar) / non-renewable fossil fuels • Materials: cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion • Wastes: biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate • Temperature: low / high (Haber process, incineration)	4

Question	Marking guidance	Marks
02.2	- Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) - They are re-used in cycles so there is no depletion of raw materials or build-up of wastes	2
02.3	- Use often disperses the minerals (eg in products spread over the world / corrosion) - Or produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced	2
02.4	- They cannot easily be removed from end-of-life appliances and are not biodegradable - Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	
03.1	- Stable tropical climate over long periods gave time for species to evolve and adapt to local abiotic and biotic factors; many rare species with small ranges - High light, warm temperatures and high rainfall all year allow continuous photosynthesis and growth, supporting rich food webs with no seasonal food shortage (no hibernation / migration) - Stratification (emergent, canopy, understorey, shrub, ground layers) and epiphytes create many niches; intense competition for light drives specialisation - Biotic pressures (predation, inter-species relationships) drive evolution of specialised pollination / seed dispersal relationships; plants flower and fruit at different times	4
03.2	- Adapted to stable conditions so they have narrow ranges of tolerance; a small change can move conditions outside the range - Many species are rare with small ranges / small populations	2
03.3	- No cold winter kills herbivores so plants defend themselves with toxic chemicals (alkaloids) that are medicinally active in controlled doses (quinine, diosgenin, curare) - Animals without teeth / claws (frogs, reptiles) rely on chemical defences or venoms (fer-de-lance venom - captopril)	2
03.4	- Over half the Amazon's rainfall comes from transpiration within the forest; clearance reduces downwind rainfall - Rapid decay recycles nutrients so soils are shallow / poor; canopy and roots prevent erosion of the thin soil - clearance leads to rapid erosion and infertility	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 52-55; AQA spec 3.1.2.3.4.	
04.1	$(240\,000 - 2\,000) / 240\,000 \times 100 = 99.2\%$	1
04.2	- Whales are k-selected: slow growing, late maturing, one calf every few years, so populations recover very slowly - Technology (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades - No regulation / open access to the ocean; whales were valuable (oil, spermaceti) and easily found	3

Question	Marking guidance	Marks
04.3	• Moratorium on commercial whaling since 1986 until populations recover • Total protection for certain species; whale sanctuaries • Limits on numbers and sizes taken; protection of suckling mothers and calves • Research into whale biology and threats (ship strikes, net entanglement); exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas	4
04.4	• ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since • Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses; whaling involves few nations and a single treaty body	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2.	
05.1	<i>Any 2 from:</i> • Hedgerows • Hay meadows • Grazed moorland / heathland • Chalk grassland	2
05.2	• Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive • Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers	2
05.3	• The introduced (non-indigenous) predator may eat other species as well as the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
06.1	• Normal diagonal mesh closes as the net fills / stretches so small fish cannot escape; square-mesh panels do not close so juveniles escape to grow and breed; TEDs let turtles out; larger mesh sizes generally • Minimum size lets fish reach breeding age before capture (eg UK lobster minimum carapace above breeding size, cod 35 cm); maximum size / V-notching egg-carrying females protects large, highly fecund breeding adults (Hebridean lobsters over 145 mm, Nassau grouper) • NTZs (Lundy, Lamlash Bay, Chagos, Great Barrier Reef) protect breeding populations and habitats; surplus young spill over and increase surrounding catches; damaged communities recover • Closed seasons, ideally in the breeding season, let fish grow and breed; fishermen may oppose because shoaling breeders are easy to catch	8
06.2	• Pelagic species form single-species shoals so fishing stops when the quota is reached • In mixed fisheries many species are caught together; once one species' quota is reached it must be discarded (dead) while fishing for others continues	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	

Question	Marking guidance	Marks
07.1	• Trees (hazel, oak) are cut to ground level every 3-30 years and allowed to regrow from the stump • About 0.5 ha cut each year on a rotation (eg 8 years) so the wood is a patchwork of areas of different ages • Each age has its own community: newly cut areas get high light for wildflowers and butterflies; older areas give dense cover for nesting birds / dormice • Pollarding cuts above browsing height so deer / livestock cannot eat the regrowth	4
07.2	• Wolves (top predator, exterminated by 1926) reduced deer populations and changed their behaviour • Fewer young trees eaten so trees and riverside vegetation recovered • More trees allowed beaver populations to increase; beaver dams increased wetland area • Wetland species (birds, fish, invertebrates) became more common; overall biodiversity rose, though wolves cause problems for livestock farmers	4
07.3	• Undesirable species (introduced predators / competitors, or natural succession species such as scrub, conifers in the New Forest heathland) reduce survival of the target species • Removal restores suitable conditions (eg heathland for sand lizards and smooth snakes) or reduces predation / competition	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3-3.1.3.4.	
08.1	• Island communities evolved from few colonising species in isolation, often without mammal predators or large herbivores • So indigenous species lack adaptations (eg ground-nesting birds, flightlessness) to survive introduced cats, rats, pigs, goats	2
08.2	• Competitor: grey squirrel (North America) out-competes red squirrel - digests acorns, larger, wins nest sites; rhododendron shades ground and releases toxins preventing native regrowth • Predator: American mink (escaped fur farms) reduced water vole range; cane toads in Australia; cats / rats / pigs on islands eating ground-nesting birds; Nile perch in Lake Victoria ate cichlids to extinction • Pathogen: squirrel pox virus carried by grey squirrels kills red squirrels; crayfish plague carried by signal crayfish kills white-clawed crayfish; Dutch elm disease, ash dieback, sudden oak death imported on plants / soil	6
08.3	• Closely related introduced species (sika deer; domestic cats) cross-breed producing fertile hybrids • The natural gene pool is changed / diluted with genes it would not naturally contain, so the pure indigenous species may disappear	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 24-25; AQA spec 3.1.2.2.	

Question	Marking guidance	Marks
09.1	• Harvest rate below the maximum sustainable yield; only economically valuable species (mahogany) selectively logged, not clear-felled • Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing; large seed trees and small trees left to grow • Buffer zones with no harvesting or machinery near rivers / wetlands • Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees • Each felled tree tagged and tracked to the retailer for FSC accreditation; rangers control illegal logging; tree nurseries replant over-exploited and ecologically important species	5
09.2	• The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it • Employment as rangers, forestry and tourism workers; ecotourists use local restaurants, produce and souvenirs; offcuts given to local carvers • Benefits give the community an incentive to protect the forest; carbon sequestration payments from companies fund management	3
09.3	• Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept; timber resource used over a longer period • Disadvantage: more labour-intensive and expensive; more tracks needed per tree	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Water: catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already; limits in arid regions • Minerals: exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging; limits: dispersal, alloys, energy of collection • Waste management: repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste; EU targets on packaging, batteries, WEEE • Pollution control: shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours); limits for legacy pollution • Energy: renewables, storage, energy density technologies, low-temperature manufacturing, CHP; limits of intermittency, transport fuels • Food: natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture; tension with maximum yield for a growing population • Land uses supporting ecosystems: urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • Judgement: high applicability in some sectors (water, waste), partial in others (minerals, energy for transport); needs policy support	9
10.2	• Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Question	Marking guidance	Marks
11.1	Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Physical features: mass / gravity retaining an atmosphere and giving pressure for liquid water; distance from the Sun and the Goldilocks zone; speed of rotation reducing temperature extremes; tilt producing seasons; magnetic field deflecting the solar wind; the Moon stabilising the orbit and producing tides • Importance of liquid water: solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • Development of life: Archaea at vents; photosynthesis; oxygen reacting with iron then accumulating; ozone layer allowing colonisation of land; aerobic organisms • Life changing conditions: carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases • Methods of studying past conditions: proxy data, isotope ratios, ice cores, Mg:Ca ratios; understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	25

Question	Marking guidance	Marks
11.2	Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Energy flow: autotrophs and heterotrophs; losses by reflection, transmission, respiration; 100 000 to 1 000 units; few food chains beyond four levels; plant food most productive; grazing animals where crops impossible; pigs on wastes • Energetics: intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • Control of genetics: asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • GM: Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape; advantages; risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity) • Sustainability links: energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	25